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Imparting boron nanosheets with ambient stability
through methyl group functionalization for
mechanistic investigation of their lithiation process

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Abstract

Although ultrahigh theoretical capacity has long been predicted for boron based lithium-ion-battery anodes, experimentally boron has only exhibited limited performance and its lithiation process remains elusive. The two dimensional (2D) form of boron is believed to be an ideal model system to investigate the lithiation behavior of boron, however unfortunately, most reported 2D boron structures are prone to oxidation under ambient

conditions. In this contribution, through a simultaneous etching and in-situ functionalization process, we synthesized for the first time methyl-functionalized boron nanosheets, which remain stable up to 250 °C. Combining experiments and theoretical calculations, we found that lithiation of boron is realized through the formation of alloys such as LiB_3 and Li_3B_{14} , while alloys with higher Li contents such as Li_5B are thermodynamically less favored. In addition, detailed electrochemical analysis reveals that side reactions on boron surface may also contribute to the unsatisfactory performance of boron based electrodes. Our findings suggest that reducing the enthalpy of formation of high Li content alloys and the choice of less nucleophilic electrolyte are the key to developing high performance anodes based on novel boron materials. Our demonstration of stable 2D boron structures also paves the way for their fundamental study and practical applications.

1. Introduction

The redox chemistries between ion charge-carriers and host materials are crucial for many electrochemical cells such as lithium-ion batteries (LIBs), sodium-ion batteries

(SIBs), lithium-sulfur batteries (LSBs) and supercapacitors.^{1,2} Graphite as a conventional intercalation-type of host material can no longer meet the ever-increasing demand for high-energy-density power sources due to its relatively small number of active sites for storing ions such as Li^+ or Na^+ .³ Consequently, host materials that operate on the basis of other mechanisms such as alloying and conversion reactions have received growing interest in the battery industry.⁴

Among the various novel anode materials explored previously,⁵ boron once attracted much attention because it can form a series of lithium alloys such as LiB_{10-13} , LiB_{6-7} , Li_3B_{14} , LiB_3 and $\text{LiB}_{0.8-1}$.^{6,7} Besides, Li-rich compounds such as Li_2B , Li_4B and Li_5B have also been predicted by simulations,⁸ and an unprecedentedly high theoretical capacity of 12395 mAh g^{-1} or 29004 mA h cm^{-3} may be achieved for forming Li_5B . This is much better than silicon (3579 mAh g^{-1} or 8375 mAh cm^{-3}),³ the most promising anode material for LIBs to replace graphite in the near future.⁴ In addition, boron is a non-toxic semiconductor with considerable low cost.⁹ Despite all of these advantages, boron has so far performed unsatisfactorily in practical cells at room temperature. For example,

Huggins et al. reported that an electrode based on crystalline boron could only uptake 0.01 mol of Li per mol of boron with poor reversibility.¹⁰ By using amorphous boron nanorods as anode, Xiong's group reported a reversible capacity of 170 mAh g⁻¹ at an extremely low current rate of 10mA g⁻¹.¹¹ Li and coworkers prepared B₅₀ thin film based anode and achieved a room-temperature capacity of 43 mAh g⁻¹, which then increased to 268 mAh g⁻¹ at 85 °C.¹²

The aforementioned deviation between theoretical predictions and experimental results can be ascribed to the uncharted lithiation process of boron.¹³ The two dimensional (2D) form of boron, i.e. atomic-layer thick boron nanosheet (e.g., borophene), may provide a model system to study this process. This is because theoretically such a 2D structure may favor ion diffusion, improve the cycle rate and overcome the volume change during charging/discharging processes.¹⁴⁻¹⁶ 2D boron and its derivatives such as borophene stabilized on different substrates and hydrogen boride nanosheets formed in solution have been reported recently.¹⁷⁻²¹ However, the strong

tendency of boron to be oxidized under ambient conditions has dramatically hindered their fundamental investigations and practical applications.

Herein, we for the first time synthesized and used methyl-terminated boron nanosheets as the model to study the lithiation process of boron. The functionalized surfaces afforded much improved oxidation resistance at room temperature and the resulting nanosheets were also dynamically stable up to 250 °C. Through a combination of experiments and simulations, we suggest that the thermodynamically preferred formation of Li_xB_y phases with low Li contents and the possible side reactions on boron surfaces which could retard Li^+ diffusion are the main reasons for the limited capacity of boron anode.

2. Experimental section

2.1 Synthesis of methyl-functionalized boron nanosheets. 0.5 g MgB_2 powder (>99%, Sigma-Aldrich), 3g iodine beads (>99%, Aladdin Chemical Reagent Co. Ltd) were mixed with 50 ml methanol ((>99.8%, ACS grade, Aladdin Chemical Reagent Co. Ltd) and stirred for 5 min to dissolve the iodine beads. The mixture was then cooled by ice bath

for 1 h before 0.5 mL 4M HCl (Technical reagent, Nanjing Chemical Reagent Co. Ltd) was dropwise added. The mixture was stirred for one week at room temperature. The product was washed with methanol four times and ultrasonic to yield isolated methyl-functionalized boron nanosheets.

2.2 Materials characterizations. X-ray diffraction (XRD, SmartLab Rigaku) was performed with Cu K α radiation ($\lambda = 1.54 \text{ \AA}$) as the X-ray source. The UV-vis-NIR absorption spectrum was recorded by LAMBDA 650S. Scanning electron microscopy (SEM, Hitachi S-4800) was used for the morphology analysis. To gain the crystal structure information, high-resolution transmission electron microscopy (HRTEM, JEOL 2100F) coupled with energy dispersive X-ray spectroscopy (EDS) analyses were performed. Electron energy loss spectroscopy (EELS) measurements were carried out using a Gatan GIF Quantum 965. Fourier Transform Infrared spectroscope (FTIR) measurement was carried out using Nicolet6700. The electrode samples for ex situ XRD were prepared by disassembling the tested half-cells after cycling at different stages. The samples were washed with dimethyl carbonate (DMC) and covered with

paraffin before the XRD characterizations. All the operation processes were carried out in the glovebox. The chemical state study of the samples were carried out by X-ray photoelectron spectroscopy (XPS, PHI 5000 VersaProbe) coupled with ultraviolet photoelectron spectroscopy (UPS) and the binding energies were corrected for specimen charging effects using the C 1s level at 284.6 eV as the reference. Thermogravimetric analysis (TGA, NETZSCH STA409) was conducted with a heating/cooling rate of 10 °C min⁻¹ between 30 to 600 °C under N₂ atmosphere.

2.3 Electrochemical measurements. The electrochemical performance of the methyl-functionalized boron nanosheets was measured with 2032 coin-type cells. Lithium metal was used as anode. The working electrode was prepared by mixing 50% of active materials, 40% of acetylene black and 10% of polyvinylidene fluoride (PVdF) in N-Methyl-2-pyrrolidone (NMP) and the slurry was coated onto a copper foil substrate. The active material loading weight is ~ 0.6 mg/cm². The calculated capacity is based on the weight of the surface functionalized boron nanosheets. 1mol L⁻¹ LiPF₆ in a mixture of ethylene carbonate (EC) and DMC (1:1 by volume) worked as electrolyte and Celgard

2300 as the separator. The charge/discharge tests were carried out between 0.01-3 V vs. Li/Li⁺ on a Neware CT-4008. Cyclic voltammetry (CV) was measured by an electrochemical workstation (Autolab 302N) between 0.01-3 V vs. Li/Li⁺ at different scan rates after cycles. The electrochemical impedance spectroscopy (EIS) measurements were carried out on an electrochemical station (Autolab 302N) at open circuit potential (OCP) condition in the frequency range of 0.1-10⁶ Hz with an AC perturbation of 5 mV in amplitude. The OCP of the pristine and cycled sample are ~2.3 V before EIS measurements. The samples were cycled 10 times with a current rate of 20 mAh g⁻¹ before CV and EIS characterizations if without specification.

2.4 Simulations. Density functional theory as implemented in the CASTEP package from Accelrys²² was used to optimize all structures and evaluate total enthalpies. We used the Perdew Burke Ernzerhof function to describe the electron-electron interaction. The structure optimization used an energy cut-off of 10 eV, k-point grids of 0.04 1/Å. The geometry optimizations and enthalpy evaluations of Li-B alloys at 1 atm were

performed and the geometries were relaxed until 0.01 eV/Å. The optimized crystal lattices were listed on Table S2.

3. Results and discussion

3.1 Characterizations of methyl-functionalized boron nanosheets. As schematically shown in Figure 1A, the methyl-functionalized boron nanosheets were prepared by selectively etching the Mg layers in MgB_2 crystals with a solution containing methanol, iodine and hydrochloric acid. Simultaneously, this mixed etchant could react to generate iodomethane slowly to realize methylation of the boron layers, which were further sonicated to yield isolated nanosheets. Figure 1B shows the SEM image of the pristine MgB_2 particles. Figure 1C and 1D show the TEM images of the boron structures after Mg removal, and the exfoliated boron nanosheets, respectively. The lamellar particles after reaction are dispersed into boron nanosheets with the thickness larger than 1 nm (Figure S1 and S2) through sonication. Owing to the nonuniformity of the MgB_2 particles, the sizes of the nanosheets distribute from less than one hundred nanometer to several micrometers (Figure S3). These boron nanosheets were further characterized by XRD

(Figure 1E). The diffraction peaks of MgB_2 disappeared after etching, and instead only broad peaks between $5\text{-}10^\circ$ were present, suggesting the lost of long-range structural order and the possible formation of thin-layered structures in the product. The absence of Mg in the EDS spectra of the product (Figure S4 and S5) implies the fully conversion of MgB_2 to boron nanosheets.

Bonding configuration and surface functionalization of the boron nanosheets were studied by EELS (Figure 2A) and FTIR (Figure 2B). The EELS spectrum confirms that B and C are the dominating species in the nanosheets, and two clear peaks shown on the B edges at 188.6 and 197.8 eV can be assigned to the excitation from B 1s to π^* and σ^* , respectively.²³ These excitation peaks indicate that the B atoms adopt a planar sp^2 bonding configuration.²¹ The FTIR spectrum shows several major modes corresponding to B–C vibration at 1050 cm^{-1} ,²⁴ $-\text{CH}_3$ rocking at 878 cm^{-1} , $-\text{CH}_3$ bending at 1240 and 1403 cm^{-1} , and $-\text{CH}_3$ stretching at 2900 and 2973 cm^{-1} .²⁵ The oxidation state of the B nanosheets was further analyzed with XPS (Figure 2C). The XPS B 1s spectrum can be deconvoluted into three major peaks, i.e. 187.3 eV, 188.4 eV and 188.9 eV. The first

two peaks are consistent with those reported previously in the B 1s spectrum of 2D boron sheets grown and stabilized on Ag(111) surface.¹⁸ The third peak at a higher binding energy of 188.9 eV is likely originated from the functionalization of boron, but not from boron oxides that are usually observed at binding energies above 190 eV.²⁰ Therefore, this peak might be a result of the methyl-functionalization of the boron sheets, in line with the FTIR result (Figure 2B). The surface functionalization has enabled stability improvement of the nanosheets. As evidently shown in their TGA result, the first transition was recorded at around 250 °C (Figure S6).

The UPS characterization of the B nanosheets in comparison with the untreated MgB₂ particles indicates an obvious change of band structure with emergence of a new peak centered ~12 eV (Figure 2D). These boron nanosheets may adopt metallic or semimetallic properties as no peaks are present in their absorption spectrum ranging from UV to near IR region (Figure S7), in agreement with previous theoretical predictions that 2D boron sheets are intrinsically metallic.²⁶⁻²⁸

3.2 Electrochemical performance of methyl-functionalized boron nanosheets. Figure

3 shows the electrochemical properties of the boron nanosheets as anode for LIBs. The voltage profiles of the first discharge-charge curve and the second discharge curve between 0.01 and 3 V at a current density of 20 mA g⁻¹ are presented in Figure 3A. At the first discharge, the potential abruptly drops to reach a plateau located between 0.8-0.9 V, similar to the previously observed discharge behavior for Li-B alloyed anode.²⁹ As the surface functionalized groups do not stop Li⁺ ions from accessing the active material,³¹ the lithiation process can be written as $\text{Li}^+ + x\text{B} + e^- \rightarrow \text{LiB}_x$, where $x \sim 3.6$ estimated from the discharge plateau in Figure 3A. Then the discharge process is dominated by the surface adsorption of Li ion owing to high surface area of the low dimensional structured anode materials and the B/Li ratio is ~ 2.8 for the discharge plateau below 0.4 V.³⁰ Similar to the functional groups on Si surface,³² the methyl group on the boron surface does affect the interfacial property of the electrode. It may form Li-contained species through reaction with electrolytes and help the formation of SEI layer. The capacity dramatically faded after the first discharge with a low reversible value of 392 mAh g⁻¹. To confirm the formation of Li-B alloy, XRD analysis (Figure 3B) was

carried out to examine the anode material before the first discharge (stage ①), after the first discharge (stage ②), and before the second discharge (stage ③). It can be seen that Li_3B_{14} and LiB_3 were formed after the first discharge process and remained in the anode for the rest of cycles. The cycling and rate performance of the boron nanosheets anode were also characterized (Figure S8 and S9)

The scan rate-dependent CV measurement (Figure 3C) of the boron nanosheet based electrode was further performed to study the charge storage kinetics. The peak current (I_p) values at near zero potential vs. Li^+/Li for different scan rates (ν) were collected for further analysis because of their large polarization. The $\log(I_p)$ vs. $\log(\nu)$ plot can be linearly fitted as shown in Figure 3D, and the slope of the plot is an indicator of the determinant factor that controls the charge storage process. It is generally accepted that a slope of 0.5 indicates a total diffusion controlled behavior, whereas 1.0 indicates a capacitive process.^{33,34} The slope of 0.61 obtained in the present study thus suggests a diffusion controlled (de)lithiation process.

3.3 The dynamic stability of Li-B alloys. Li-B alloys with high Li contents (~70 at%) have been employed previously as reversible negative electrodes in molten salt batteries at 400-600 °C.³⁵ However, less than 25 at% of Li content was achieved in the present study, and even lower Li contents were observed in previously reported boron based electrodes operating at room temperature.^{3,10,11} In order to understand the origin of the contradictory observations, a dynamic stability study of the Li_xB_y alloys was carried out based on the density functional theory (DFT, Figure 4 and Figure S10). Figure 4A illustrates the ground-state crystal structures of the related Li_xB_y phases and figure 4B shows the relative ground-state enthalpies of formation (ΔH_f) for the various Li_xB_y phases with respect to the elemental crystals of Li and B. As ΔH_f reflects the stability of the binary phases, it can be seen that Li_3B_{14} and LiB_3 are dynamically much more stable within the calculated Li_xB_y phases. This is consistent with our experimental observations (Figure 3B). This simple simulation presented one possibility that the conversion of LiB_3 to Li_xB_y phases with higher Li contents, such as LiB and Li_5B , was unlikely to occur at room temperature.

3.4 The kinetics of the lithiation process. Apart from the formation of Li_xB_y phases with low Li contents, the sluggish Li^+ diffusion-controlled kinetics evidenced in Figure 3C-D is another cause to the low performance of the boron anode. To understand the origin of the sluggish diffusion process, we first estimated the activation energy (E_a) of Li^+ diffusion via EIS measurements. As shown in Figure 5A, the boron electrode before cycling exhibited an increased charge transfer resistance (R_{ct}) with decreasing temperature. The activation energy (E_a) of Li^+ transport between the electrolyte and anode interface can thus be calculated based on the $\ln(1/R_{ct})$ vs. $1000/T$ plot in Figure 5B.³⁶ An E_a value of 473 mV is obtained which is only slightly larger than that of the conventional carbon anode (~400 mV).³⁷ After activation via cycles of (de)lithiation, the electrode impedance doubled due to the formation of solid electrolyte interphase (SEI) layer (R_{SEI}), and the much increased charge transfer resistance (R_{ct}) based on the fitting of the equivalent circuits (Figure 5C and Table S1). This also correlates to a much larger E_a of 637 mV (Figure 5D). The increased R_{ct} can be a result of the hindered Li^+ transportation within the boron electrode or between the electrode and SEI. However, based on the EIS results, the diffusion coefficient of Li^+ (D_{Li}) within the boron anode

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3 before and after cycling was both estimated as $\sim 10^{-8} \text{ cm}^2\text{s}^{-1}$ (see the note in SI), which
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7 is on the same level with that of graphite (10^{-9} - $10^{-7} \text{ cm}^2\text{s}^{-1}$).³⁸ Therefore, the slow
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10 kinetics observed in the boron based electrode is likely arising from the restricted Li^+
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13 diffusion from the SEI layer to boron, rather than within the boron layer itself.
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18 To reveal whether unwanted electrochemical events occurred between the SEI and
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21 boron electrode, a galvanostatic-intermittent-titration technique (GITT) analysis was
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24 employed (Figure 6A). Although the time constant for Li^+ diffusion ($\sim 30 \text{ s}$, estimated
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27 from equation $L = \sqrt{Dt}$, where L is the diffusion length and D is the diffusion coefficient)
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30 is much smaller than the relaxation time (10 min) set for the GITT analysis, an
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33 equilibrium potential was not obtained under the relaxation condition (Figure 6A-B).
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36 Therefore, open-circuit potentials (OCPs) were monitored by relaxing a cell after its
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39 being lithiated from 3V to 1V (red curve) or delithiated from 0.01V to 1 V (black curve)
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43 for 48 h (Figure 6C). For both curves, sharp potential variation was observed within the
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47 first 2 h, after which the increase of the OCPs became much slower but did not stop
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51 even at end of the experiment. This scenario points to the possible presence of side
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reactions at the interface of SEI/electrode,³⁹ or the non-equilibrium dynamics of the boron electrode. To eliminate the effect from the latter, the OCP curve was further analysed by a methodology established by Davis et al. and amended by Sethuraman et al.^{40,41} The relaxed OCP without considering side reactions can be written as follows:

$$V = U + \frac{RT}{\alpha_c F} \ln \left\{ \frac{\alpha_c F i_0}{RT C_{dl}} t + \exp \left[\frac{\alpha_c F}{RT} (V_0 - U) \right] \right\} \quad (1)$$

Where U is the equilibrium potential, α_c is the cathodic transfer coefficient, i_0 is the rate constant, V_0 is the initial potential, C_{dl} is the double-layer capacitance and F is Faraday constant.

For long relaxation times, the relation can be simplified as follows:

$$\frac{dV}{d \ln t} = \frac{RT}{\alpha_c F} \quad (2)$$

It shows a linearly proportional relationship between the OCP and $\ln t$ at long relaxation times without a side reaction. As shown in Figure 6D, an α_c value of 0.47 is estimated from the above equation based on the red curve in Figure 6C. This value is close to 0.5, indicating the main (de)lithiation reaction is in a state of dynamic

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3 equilibrium.⁴¹ Therefore, the observed non-equilibrium relaxation in Figure 6C and the
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7 nonlinearity of the curve in Figure 6D both can be attributed to the presence of side
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10 reactions.

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15 As elemental boron is electron deficient and can easily coordinate with nucleophilic
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18 functional groups like carbonyl group,⁴² the side reaction may arise from the interaction
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21 between the boron electrode and the organics in the SEI layer such as $(\text{CH}_2\text{OCO}_2\text{Li})_2$
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24 which is usually formed via reduction of the electrolyte such as EC.⁴³ This can lead to a
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27 change of the chemical state of the surface boron. The XPS B 1s peaks of the electrode
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30 showed a shift to higher energies after lithiation by comparing them with those before
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33 lithiation (Figure S11). Besides, new peaks were emerged at ~ 192 eV, which can be
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36 attributed to the interaction between boron and oxygen radicals from the organics.⁴⁴
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42 These peaks became much weaker when the surface layer of the electrode was in-situ
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45 removed via Ar ion sputtering during XPS measurement, confirming that organics
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48 present in the SEI reacted with the surface layer of the boron electrode. Thus-formed
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51 boron based coordination compounds could passivate the electrode surface and retard
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the diffusion of Li^+ ions, which leads to the poor lithiation/delithiation reversibility of the anode under the applied current density. As a control experiment, the original electrolyte of EC and DMC which can generate nucleophilic $(\text{CH}_2\text{OCO}_2\text{Li})_2$ during SEI formation, was replaced by a mixture of 1,3-dioxolane (DOL) and dimethoxy ethane (DME) (v:v=1:1) which mainly generates $\text{CH}_3\text{CH}_2\text{OCH}_2\text{OLi}$, a much less nucleophilic product than $(\text{CH}_2\text{OCO}_2\text{Li})_2$.^{45,46} By changing the electrolyte, the resulting half-cell exhibited an enhanced electrochemical performance in terms of specific capacity and charge transfer resistance after cycling (Figure S12-13). This further implies that the formation of boron based coordination compounds on the electrode surface may retard Li^+ diffusion.

4. Conclusions

In summary, we synthesized surface-functionalized boron nanosheets with enhanced environmental stability. These nanosheets provide a reliable model system for the fundamental study and practical application of boron-based 2D materials. Particularly, they have been used to investigate reasons behind the huge deviation between the experimental and theoretically predicted performance of boron based

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3 anode for LIBs. This deviation is believed to be attributed from the thermodynamically
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6 preferred formation of Li_xB_y phases with low Li contents and the possible side reactions
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9 on boron surfaces which could retard Li^+ diffusion. As the above issues may be
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12 alleviated through reducing the enthalpy of formation of the Li-rich phases by elemental
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15 doping ¹³ and choosing the less nucleophilic electrolytes, high performance boron-based
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18 electrode materials may be realized in future.
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25 ASSOCIATED CONTENT

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30 **Supporting Information.** The Supporting Information is available free of charge on the
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33 ACS Publications website.
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38 TEM image of the CH_3 -terminated boron nanosheet (Figure S1), AFM image of the
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41 methyl-terminated boron nanosheet (Figure S2), SEM image of the methyl-
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44 functionalized boron nanosheets (a) and the enlarged image (b) shows the nanosheets
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47 were bundled and folded together on the substrate (Figure S3), EDS mapping of a
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50 typical methyl-functionalized boron nanosheet(Figure S4), EDS spectrum and element
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53 concentration of methyl-functionalized boron nanosheets(Figure S5), TGA of the
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3 methyl-functionalized boron nanosheets in flowing N₂ gas (Figure S6), UV-vis-NIR
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7 spectrum of methyl-functionalized boron nanosheets dispersed in methanol (Figure S7),
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10 Cycling performance of the boron nanosheet based electrode at current density of 20
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13 mA g⁻¹(Figure S8), Rate performance of the boron nanosheets with different
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17 charge/discharge currents (Figure S9), Crystal models of (a) B (space group: P4₂/nnm)
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20 and (b) Li (space group: Im-3m) used as references for numerical calculation of
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24 enthalpies of formation for various Li_xB_y phases based on DFT (Figure S10), XPS B 1s
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27 spectra collected from as-prepared boron nanosheets (dark), the electrode based on
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31 boron nanosheets after cyclic lithiation/delithiation and rinsing with DMC (blue), and the
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35 rinsed electrode further treated with Ar ion sputtering etching to remove SEI (red)
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38 (Figure S11), Specific capacities of the boron nanosheet based electrode in EC/DMC
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42 electrolyte and DOL/DME electrolyte, respectively. It can be seen that the performance
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46 of the boron electrode in DOL/DME is better than in EC/DMC (Figure S12), Room
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49 temperature Nyquist plots of the boron nanosheet based electrode in EC/DMC
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53 electrolyte and DOL/DME electrolyte, respectively. The latter shows much smaller
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57 charge transfer resistance (Figure S13), Resistance data of the boron anodes before
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and after cycling, obtained by fitting the EIS in Figure 5 based on the equivalent circuits shown as insets in Figure 5 (Table S1), Crystal structure parameters of Li_xB_y phases (Table S2).

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Author Contributions

S.L. and X.H. guided the research direction and proposed the ideas. S.L., Q.Z and H.Z. carried out the synthesis experiments. Q.Z., H.Z., B.C and H.L. carried out the characterizations. Q.Z. fabricated the half-cell batteries and did all the electrochemical characterizations. S.L. performed the simulations. S.L and X.H. wrote the manuscript, and all authors contributed to the modification of the manuscript.

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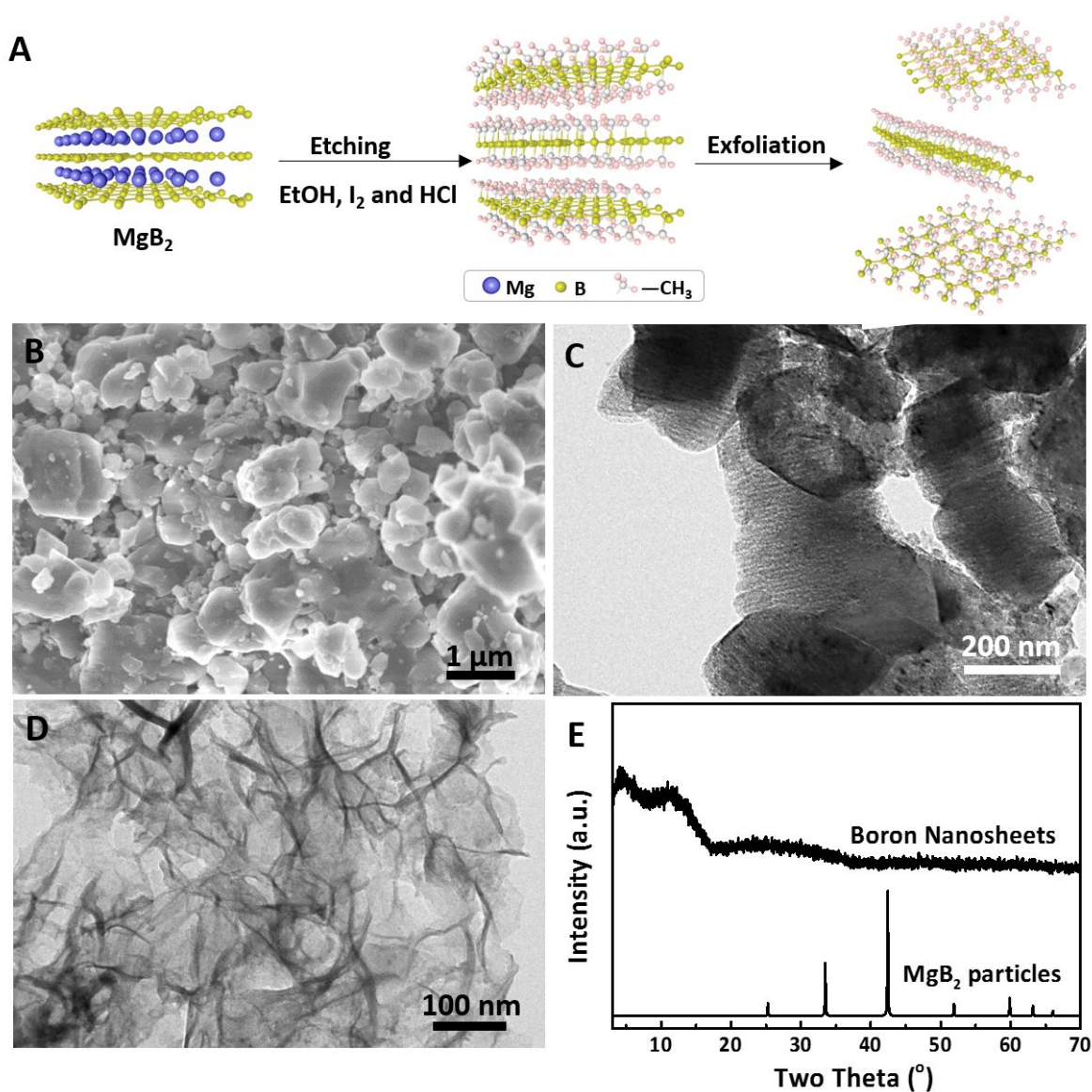


Figure 1. (A) Schematic illustration of the preparation process of methyl-terminated boron nanosheets. (B) SEM image of the pristine MgB_2 particles. (C) TEM image of the particles after Mg removal. (D) TEM image of the methyl-terminated boron nanosheets. (E) XRD patterns of MgB_2 particles and the isolated boron nanosheets.

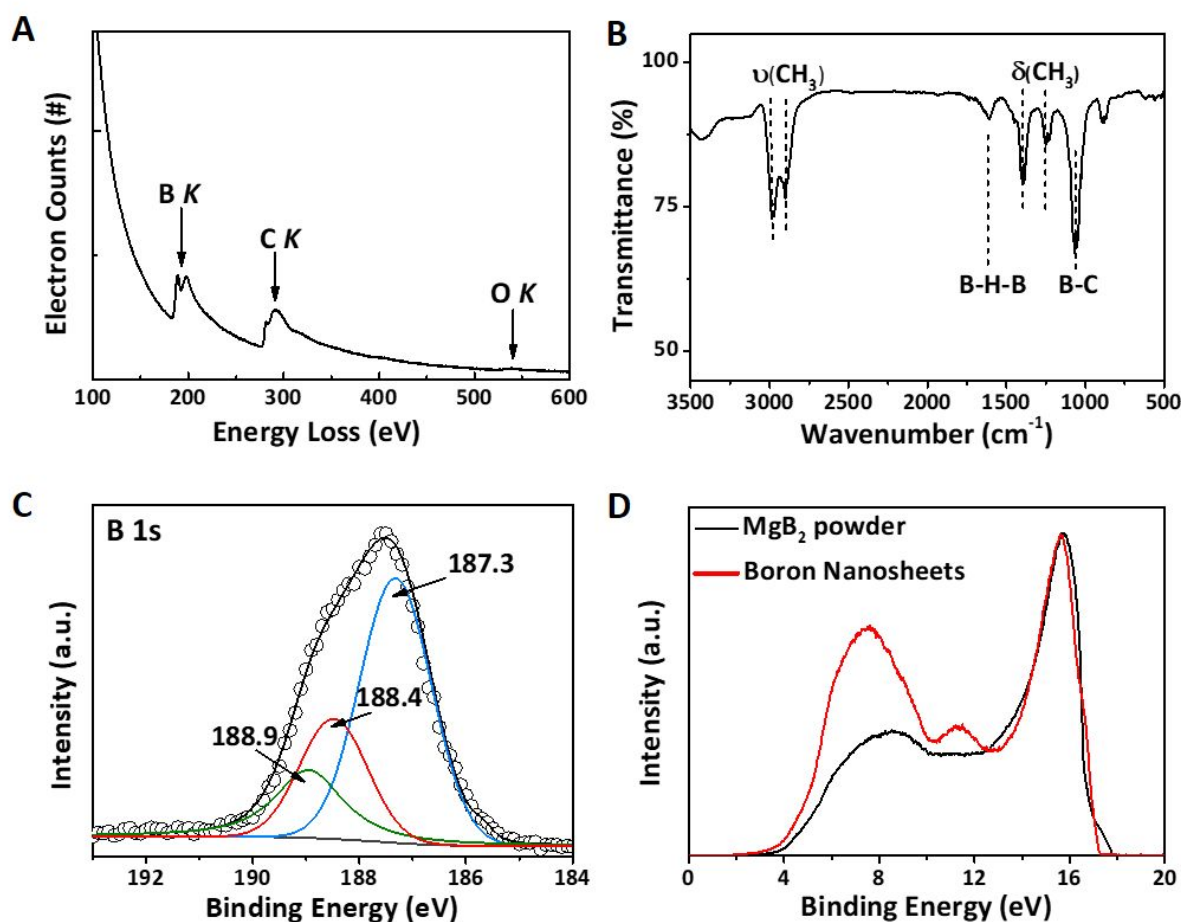


Figure 2. (A) EELS spectrum, (B) FTIR spectrum, and (C) XPS B1s spectrum of the methyl-terminated boron nanosheets. (D) UPS of the pristine MgB₂ particles and the methyl-terminated boron nanosheets; the intensity of spectra was normalized based on the peak centered at 16 eV.

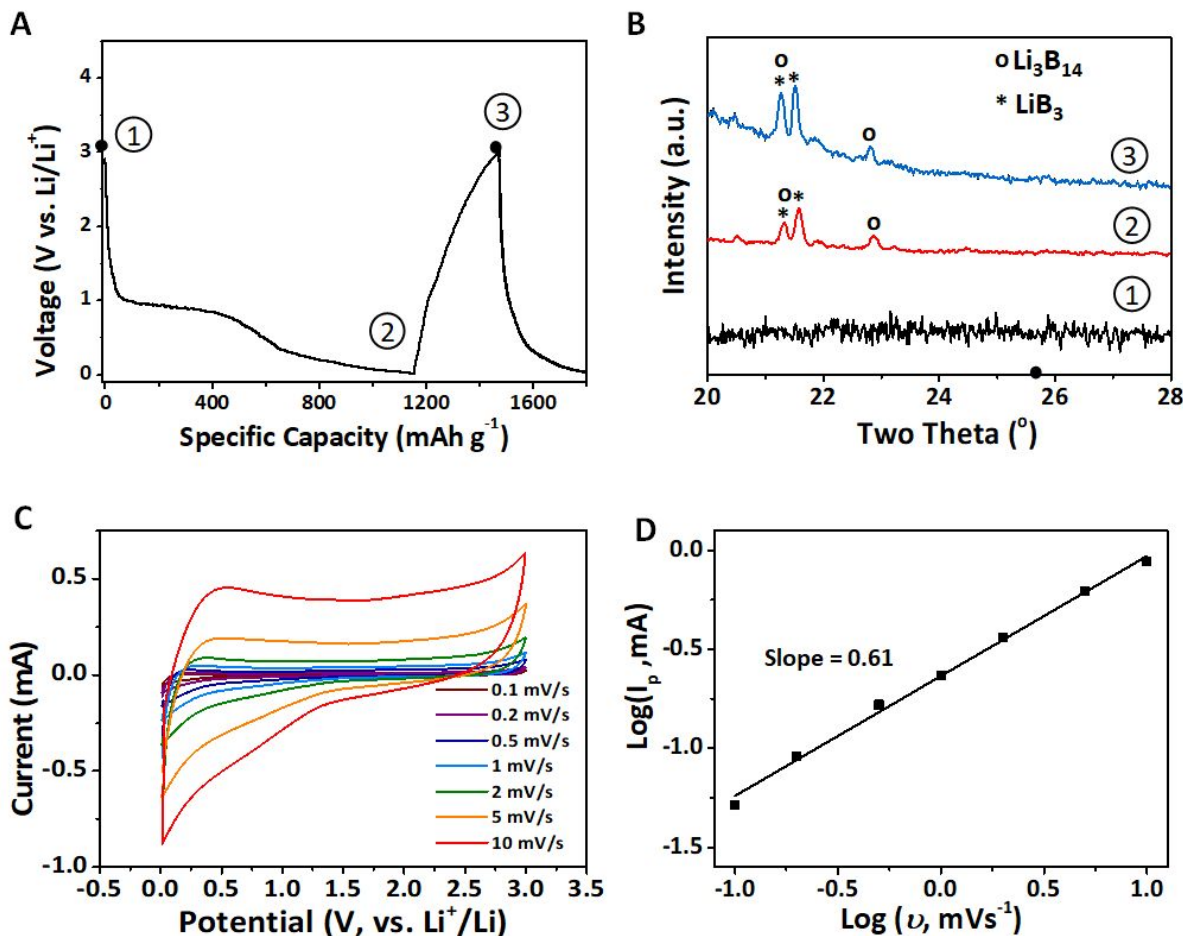


Figure 3. (A) The voltage profile of the boron nanosheet based electrode. (B) XRD patterns of the electrode material at various stages of the charge/discharge process, denoted by ①, ② and ③ in (A). (C) CV curves of the boron anode at different scan rates. (D) Linear fitting of the $\log(I_p)$ vs. $\log(v)$ plot from the CV curves in (C).

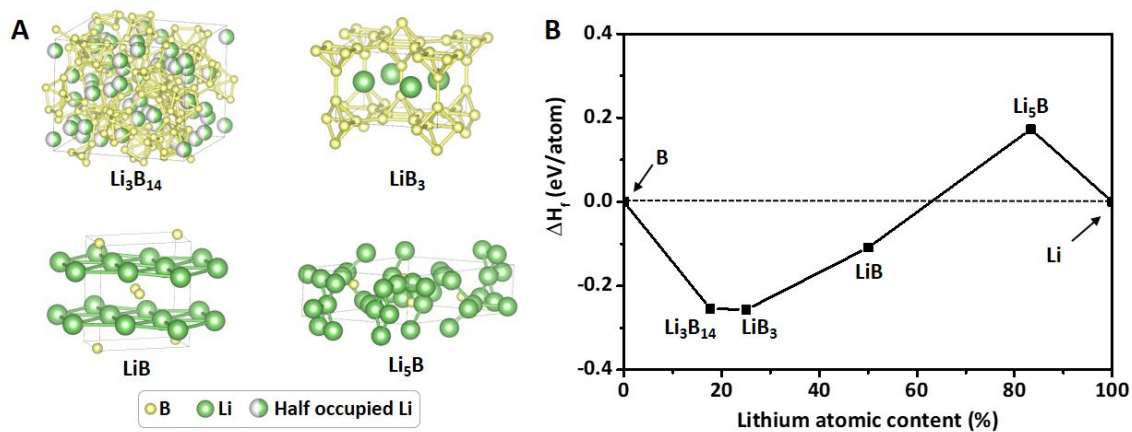


Figure 4. (A) Ground-state crystal structures of Li_3B_{14} , LiB_3 , LiB and Li_5B . (B) the relative ground-state enthalpies of formation (ΔH_f) for the various Li_xB_y phases with respect to the elemental crystals of Li and B.

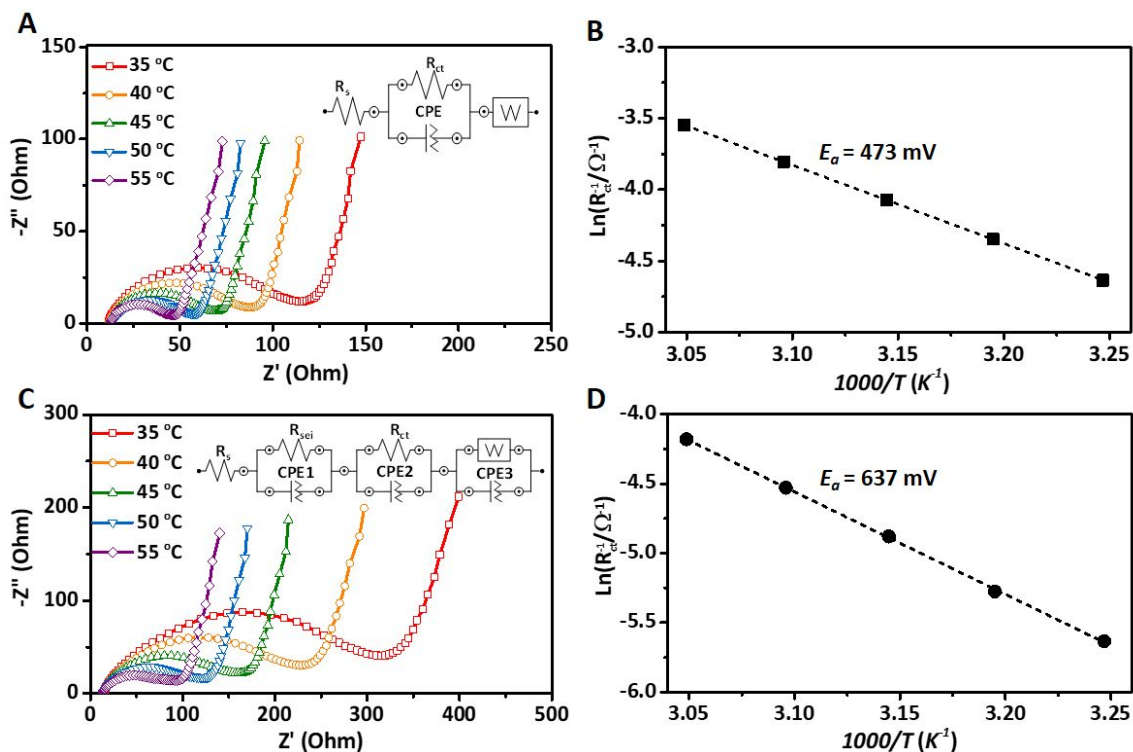


Figure 5. (A) Temperature-dependent Nyquist plots of the as-prepared boron nanosheet anode before cycling; inset is the equivalent circuit. (B) Arrhenius behavior of R_{ct} and E_a calculated based on the slope of the plot of $\ln(1/R_{ct})$ vs. the inverse of temperature (T). (C) temperature-dependent Nyquist plots of the boron nanosheet anode after cycling; inset is the equivalent circuit. (D) Arrhenius behavior of R_{ct} and E_a calculated based on the slope of the plot of $\ln(1/R_{ct})$ vs. the inverse of temperature (T).

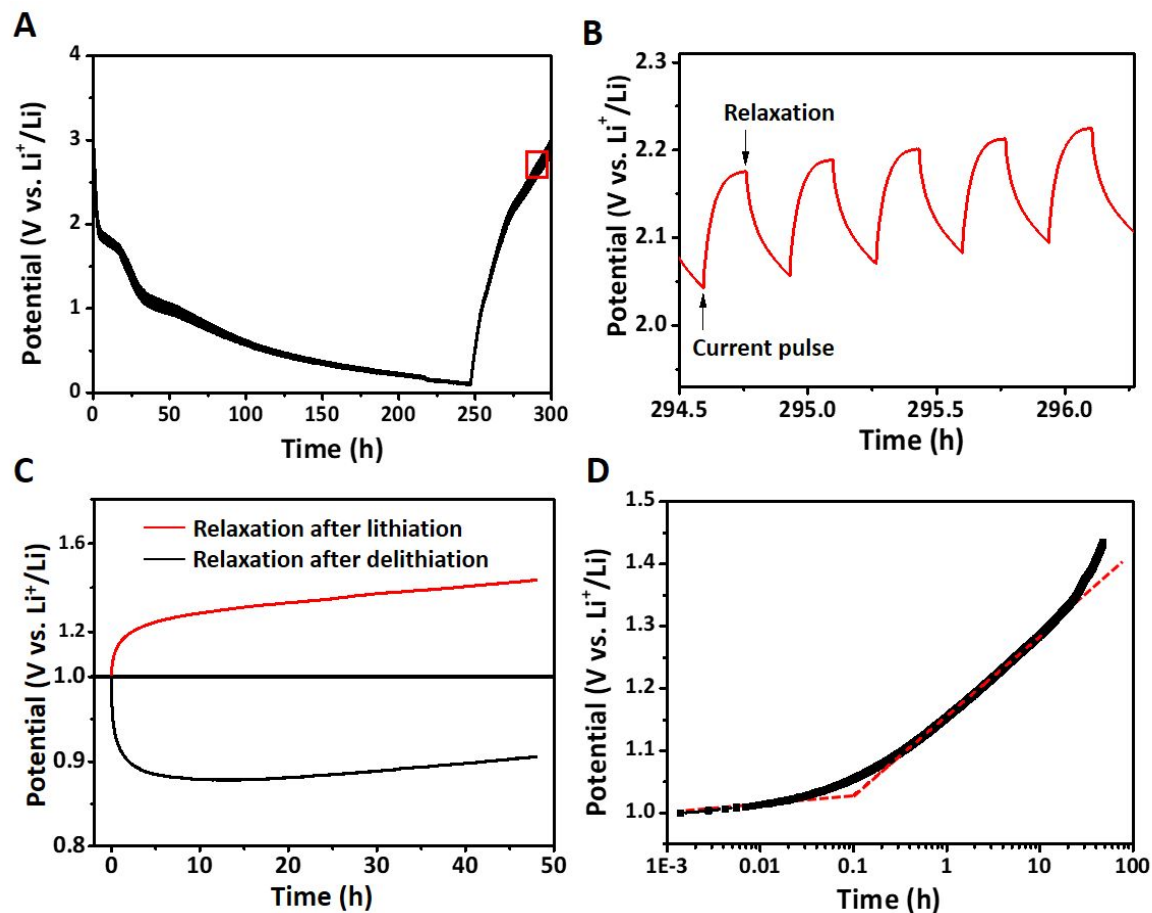


Figure 6. (A) GITT for the as-prepared cell obtained at a current density of 10 mAh g^{-1} . (B) the enlarged GITT curve obtained at $\sim 295 \text{ h}$, as highlighted in the red rectangle in (A). (C) OCPs of relaxation of a cell after its being lithiated from 3 V to 1 V at a constant current of 20 mA g^{-1} (red curve) or delithiated from 0.01 V to 1 V at a constant current of 20 mA g^{-1} (black curve) for 48 h . (D) OCP vs. $\ln t$ plot obtained from the red curve in (C); and linear Marquardt-Levenberg fitting of the curve (red dash line) without considering any side reactions.

TOC: Methyl-functionalized boron nanosheets with enhanced environmental stability

have been synthesized.

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